Unconfoundedness

BUSS975: Causal Inference in Financial Research

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Acknowledgments and readings

- These slides follow Scott Cunningham, *Causal Inference: The Remix* (2nd ed.), Chapter 5, "Unconfoundedness."
  - Free online: `https://mixtape.scunning.com`
- The Titanic example, the covariate taxonomy, and the NSW discussion are Cunningham's.
- The comparison of matching with regression, and the discussion of what matching cannot do, are adapted from Todd Gormley's lecture notes.

Outline

1. Introduction: what actually resolves bias
2. Confounders, covariates, and colliders
3. Identification assumptions
4. Subclassification
5. Exact matching
6. Nearest-neighbour matching
7. Inference with matching
8. Bias correction
9. Propensity scores
10. Regression
11. What matching cannot do
12. The NSW demonstration

Introduction

Where we left off

- Lecture 2 gave us a rule for **which** variables to condition on: block every backdoor path, condition on no descendant of D.
- It was silent on two things:
  - What if the variables that would close the backdoors are **unobserved**? (Usually the case.)
  - Given a valid set, **how** do we actually condition — and does it matter?
- This lecture takes the optimistic branch: **suppose the confounders are observed**. What then?
- The methods — subclassification, matching, propensity scores, regression — are all answers to "how." They differ less than their names suggest.

Treatment assignment mechanisms resolve bias, not models

Recall Lecture 1's decomposition. It is an **identity** — true in every dataset, experimental or not:

$$\underbrace{E[Y|D{=}1] - E[Y|D{=}0]}_{SDO} = ATE + \underbrace{E[Y^0|D{=}1] - E[Y^0|D{=}0]}_{\text{selection on levels}} + \underbrace{(1-\pi)(ATT - ATU)}_{\text{selection on gains}}$$

- Nothing here is an assumption. The identity holds whatever you do.
- **A regression cannot delete the last two terms.** No functional form, no set of fixed effects, no clustering choice touches them.
- What deletes them is a **claim about how treatment was assigned**.

What randomization actually did

- Under $(Y^1, Y^0) \perp\!\!\!\perp D$, we were permitted to *deduce*:

$$E[Y^0|D{=}1] = E[Y^0|D{=}0], \qquad E[Y^1|D{=}1] = E[Y^1|D{=}0]$$

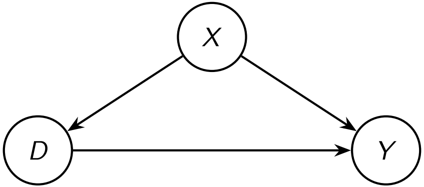
- The first kills selection on levels. Both together kill $ATT - ATU$. What is left is the ATE.
- The deductions came from the **assignment mechanism**, not from any estimator. We then *inserted* them into a statistical model.

The statistical model alone cannot resolve bias — the statistical model matched with the assignment mechanism can.

Why this framing matters

- Randomization is an **assignment mechanism**, not a statistical model. A regression is a **statistical model**, not an assignment mechanism.
- Confusing the two produces the most common error in empirical finance: believing that a long enough control list is itself an identification strategy.
- Unconfoundedness is this lecture's assignment mechanism. It says treatment was assigned **as if at random within strata of** X.
- Everything that follows — subclassification, matching, propensity scores, regression — is a statistical model that becomes credible *only* if you can defend that mechanism.
- So the order of business is: defend the mechanism first, choose the estimator second.

The bet we are making



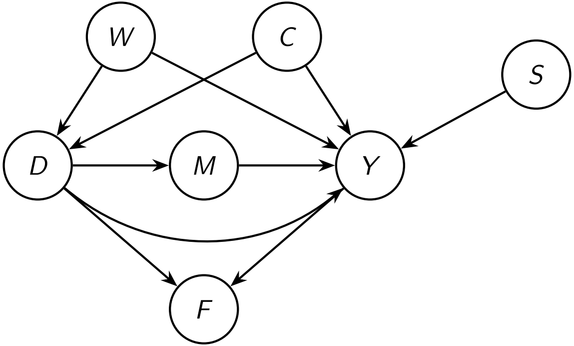
- Everything rests on one claim: **there is no dashed circle**. No unobserved U pointing at both D and Y.
- That claim is **not testable**. No diagnostic, balance table, or specification test can establish it.
- It is defended with institutional knowledge, not statistics — which is why this lecture spends as much time on *what could go wrong* as on estimators.
- The literature calls it **unconfoundedness**, **conditional independence (CIA)**, or **selection on observables**. Same thing.

Confounders, Covariates, and Colliders

A worked DAG: first class on the Titanic

- **Question**: did a first-class ticket (D) raise your chance of surviving (Y)?
- The raw comparison is stark: 62.5% of first-class passengers survived, versus 27.1% of everyone else — a gap of 35.4 percentage points.
- But the *"women and children first"* rule was actually enforced, and first-class cabins were closer to the boat deck.
- So **sex** and **age** drive both who bought a first-class ticket and who got into a lifeboat. Textbook confounding.
- We use this because the data are real, complete, and small enough to compute by hand — and because the confounders are not in dispute.

The Titanic DAG



D first class Y survival W sex C age M lifeboat F fame S swimming

- Backdoor paths: $D \leftarrow W \rightarrow Y$ and $D \leftarrow C \rightarrow Y$. Both **open**. Condition on $\{W, C\}$ and both close.
- F is a **collider**: $D \rightarrow F \leftarrow Y$. Already closed; conditioning on it *opens* it.
- M is a **mediator**: first class sat nearer the boat deck. It is *part of* the effect, not a confounder of it.

Four kinds of variable

Type	Example	What conditioning does
Confounder	W sex, C age	**Required.** Closes a backdoor path.
Collider	F fame	**Forbidden.** Opens a closed path.
Neutral / precision	S swimming	Optional. No bias either way; shrinks the standard error.
Mediator	M lifeboat	**Forbidden** for the total effect. Blocks the causal path.

- Only one of these four rows is about bias *reduction*. Two are about bias *creation*.
- Note S: it predicts the outcome but not treatment. Free precision — the honest case for controls that are not confounders.
- M is the subtle one. Conditioning on it asks: *did first class help, holding fixed the main way it helped?* Near zero — and the wrong question.
- The categories are properties of the **graph**, not the variable. Press coverage *before* boarding would not be a collider.

The same DAG in corporate finance

Effect of a **bank credit line** (D) on **firm investment** (Y):

Titanic	Finance analogue	Rule
W, C sex, age	Firm size, industry, credit rating	Condition on
F fame	*Subsequent* analyst coverage	Never condition on
S swimming	Pre-period export-market demand	Optional, improves precision
M lifeboat	Amount actually drawn on the line	Never condition on
—	Unobserved managerial quality	The thing that sinks you

- Analyst coverage is the finance collider you will meet: it responds *both* to the financing event and to firm performance.
- The drawn amount is the mediator a referee will *ask* you for — "surely control for how much they actually borrowed." That removes the channel you are trying to measure.
- The last row has no Titanic counterpart, and that is the whole problem. On the Titanic we know the confounders are sex and age. In finance we rarely do.

Covariate selection without a DAG

In practice you rarely have expert consensus on the full graph. Three working rules:

1. **Include variables you are confident are confounders.** Prior knowledge about the assignment mechanism is the input, not the data.
2. **Include pre-treatment variables that strongly predict the outcome.** These are either confounders or precision-improvers like S. Crucially, they are *probably not colliders* — a variable measured before treatment cannot be caused by it.
3. **Exclude anything that might be an outcome.** If you are unsure, exclude everything measured after treatment.

- Rule 2 does most of the work in practice, and rule 3 is the one most often violated in finance.
- "Pre-treatment" is doing heavy lifting in rule 2. It is a timing argument, and timing is checkable.

Identification Assumptions

Selecting the target parameter

Before estimating anything, decide *whose* effect you want.

$$ATE = E[Y^1 - Y^0], \quad ATT = E[Y^1 - Y^0 \mid D{=}1], \quad ATU = E[Y^1 - Y^0 \mid D{=}0]$$

- Cunningham's example: a polio vaccine. The three parameters answer three different policy questions.
  - ATE: what if we vaccinate **everyone**?
  - ATT: how much did the vaccine help **those who took it**?
  - ATU: what would happen if we reached **the currently unvaccinated**?
- These coincide only under constant treatment effects. With heterogeneity they differ, sometimes a lot.
- **The estimand is a choice you make about the policy question.**

Which parameter does the question need?

Finance question	Parameter
Should the regulator extend this mandate to all firms?	*ATE*
Did the firms that adopted IFRS benefit from it?	*ATT*
Would forcing non-adopters to adopt help them?	*ATU*

- Note the third row: it is the *expansion* question, and it is almost never the one your data can answer well.
- Papers routinely estimate the *ATT* and then write policy conclusions that require the *ATE* or *ATU*. Watch for this in referee reports — including your own.

Unconfoundedness is a version of independence

- **Strong unconfoundedness:**

$$(Y^1, Y^0) \perp\!\!\!\perp D \mid X$$

- Compare Lecture 1's randomization assumption, $(Y^1, Y^0) \perp\!\!\!\perp D$, with no conditioning.
- In words: among units with the *same* X, treatment is as good as randomly assigned. It is **conditional randomization**.
- This is the assignment mechanism from §1, stratum by stratum:

$$E[Y^1 \mid X{=}x] = E[Y^1 \mid D{=}1, X{=}x] = E[Y^1 \mid D{=}0, X{=}x]$$

$$E[Y^0 \mid X{=}x] = E[Y^0 \mid D{=}1, X{=}x] = E[Y^0 \mid D{=}0, X{=}x]$$

- Strong because it's conditional independence with respect to **both** potential outcomes.

What the assumption buys: plug-in

- The fundamental problem of causal inference is that Y^1 and Y^0 are never both observed.
- Unconfoundedness lets us **replace a missing counterfactual with an observed outcome** from a comparable unit:

$$\underbrace{E[Y^0 \mid D{=}1, X{=}x]}_{\text{never observed}} = \underbrace{E[Y^0 \mid D{=}0, X{=}x]}_{\text{observed}} = \underbrace{E[Y \mid D{=}0, X{=}x]}_{\text{in your data}}$$

- The first equality is the **assumption**. The second is the switching equation — pure definition.
- Every method in this lecture is a different way of executing that substitution.
- So switching estimators cannot rescue a design: all of them lean on the first equality, and none of them can test it. **Identification** credibility is common to all.
- They *do* differ downstream — functional form, extrapolation into thin regions, which strata get weight. Those are **estimation** differences; §11 returns to them.

Unconfoundedness is controversial

- Read literally, the assumption says: firms with identical observables effectively **flip a coin** over a major financial decision.
- Every model of optimizing behaviour says otherwise. Firms choose treatment *because* they expect to benefit — Lecture 1's selection on gains.
- This is the deepest objection, and it comes from economics, not statistics.
- It is not a reason never to use these methods. It is a reason to state, explicitly, why *your* setting escapes it.

Guido W. Imbens's three defences

1. **Statistical motivation.** It is the natural starting point.
2. **It is an empirical question.** With enough institutional knowledge, the relevant X may genuinely be measurable. Whether it is depends on the setting, not on econometrics.
3. **Objectives differ.** The decision-maker optimizes over something *other than your outcome*. A firm choosing a credit line optimizes liquidity and covenant slack — not the investment measure you happen to study.

- Defence 3 is the strongest available, and worth naming explicitly when you use it.
- It also tells you when the design *fails*: when the agent is optimizing over precisely your outcome variable.

Road map: two routes through this chapter

Everything ahead assumes unconfoundedness. The methods split on **what they do when a treated unit has no comparable control.**

	Route 1: interpolation	**Route 2: extrapolation**
Rests on	Common support	Functional form
Fills the gap by	It does not — it refuses	Projecting the fitted model into it
Methods	Subclassification, matching, propensity scores	Regression
Sections	§4–§9	§10

Common support

- Unconfoundedness alone is not enough. You also need units to compare.
- **Strong common support** (needed for the ATE):

  $$0 < \Pr(D = 1 \mid X = x) < 1 \quad \text{for all } x$$

  At every value of X, both treated and untreated units must exist. <!– - Cunningham's jigsaw analogy: **unconfoundedness is the blueprint; common support is whether the pieces are actually in the box.**
- A perfect blueprint is useless if a piece is missing — and a complete box is useless if the blueprint is wrong. –>

The two assumptions differ in kind

Unconfoundedness	**Common support**
Untestable	**Verifiable in your data**
Defended with institutional knowledge	Checked with a histogram
Failure $\Rightarrow$ bias	Failure $\Rightarrow$ extrapolation

- You can have one without the other, and the failures look nothing alike.
- Common support is the assumption you have **no excuse** for not checking — and the one most papers skip.
- This distinction is the key to the NSW debate at the end of this lecture.

Weighting treatment effects under unconfoundedness

Put the two assumptions together and the *ATE* becomes computable.

$$
\begin{aligned}
ATE(x) &= E[\,Y^1 \mid X{=}x\,] - E[\,Y^0 \mid X{=}x\,] \\
&= \underbrace{E[\,Y^1 \mid D{=}1, X{=}x\,] - E[\,Y^0 \mid D{=}0, X{=}x\,]}_{\text{unconfoundedness — used on } \textit{both} \text{ terms}} \\
&= \underbrace{E[\,Y \mid D{=}1, X{=}x\,] - E[\,Y \mid D{=}0, X{=}x\,]}_{\text{switching equation}}
\end{aligned}
$$

$$
ATE = \underbrace{\sum_x \Big(E[Y \mid D{=}1, X{=}x] - E[Y \mid D{=}0, X{=}x]\Big)\Pr(X{=}x)}_{\text{iterated expectations; common support}}
$$

- $ATE(x)$ is the effect **within stratum** x — *conditional average treatment effect* (CATE). The last line averages those with **population** weights.
- **Unconfoundedness is used twice**, once for Y^1 and once for Y^0. That is precisely why the *ATE* needs the *strong* version.

A three-stratum example

Suppose the effect on mortality differs by age group, and the groups are equally sized:

Stratum	Effect in stratum (pp)	$\Pr(X = x)$
Young	80	$1/3$
Middle-aged	55	$1/3$
Old	100	$1/3$

$$\begin{aligned}\widehat{ATE} &= \frac{N_y}{N} \times SDO_y + \frac{N_m}{N} \times SDO_m + \frac{N_o}{N} \times SDO_o \\ &= \frac{1}{3} \times 80 + \frac{1}{3} \times 55 + \frac{1}{3} \times 100 \\ &= 78.33\end{aligned}$$

- The ATE is a number **no stratum experiences**. It describes a population, not a unit.

Estimating the ATT under weaker assumptions

- A weaker assumption suffices if you only want the ATT:

$$Y^0 \perp\!\!\!\perp D \mid X$$

- Only the **untreated** potential outcome must be independent of treatment.
- It allows firms to *anticipate their own gains* Y^1 and select on them, while assuming they do not select on how they would have done anyway.
  - In finance that ordering is often defensible. A firm knows its expected benefit from a credit line better than it knows its counterfactual path without one.
- **Weak common support**: Controls must exist wherever treated units do — but not conversely.

$$\Pr(D = 0 \mid X) > 0 \quad \text{for all } X \text{ such that } \Pr(D = 1 \mid X) > 0$$

Why the asymmetry makes sense

$$
\begin{aligned}
ATT(x) &= E[Y^1 \mid D{=}1, X{=}x] - E[Y^0 \mid D{=}1, X{=}x] \\
&= \underbrace{E[Y \mid D{=}1, X{=}x]}_{\text{switching equation}} - E[Y^0 \mid D{=}1, X{=}x] \\
&= E[Y \mid D{=}1, X{=}x] - \underbrace{E[Y \mid D{=}0, X{=}x]}_{\text{weak unconfoundedness}}
\end{aligned}
$$

$$
ATT = \underbrace{\sum_x \Big(E[Y \mid D{=}1, X{=}x] - E[Y \mid D{=}0, X{=}x]\Big) \Pr(X{=}x \mid D{=}1)}_{\text{common support, wherever treated units live}}
$$

- Weighting by $\Pr(X \mid D{=}1)$ — covariates **among the treated** — is what makes it the ATT: strata with no treated units get zero weight.
- **Rule**: if you can only defend the weak version, your estimand is the ATT. Say so.

Subclassification

Cochran's method

- The oldest estimator here, and the most transparent. Three steps:
  1. **Stratify** on X.
  2. Compute the treated-minus-control difference **within** each stratum.
  3. **Average** the strata, weighting by whatever the estimand requires.
- No functional form is assumed. Within a stratum, units are identical on X by construction.
- The weights define the estimand:

Estimand	Weight stratum k by	Interpretation
ATE	n_k/N	population share
ATT	$n_{k,\text{treated}}/N_{\text{treated}}$	treated share
ATU	$n_{k,\text{untreated}}/N_{\text{untreated}}$	control share

The Titanic: the raw comparison

- Treated: 325 first-class passengers, 62.5% survived.
- Control: 1,876 others, 27.1% survived.

$$SDO = 62.5 - 27.1 = 35.4 \text{ percentage points}$$

- Taken at face value: a first-class ticket nearly *tripled* your survival odds.
- But the first-class cabins held a very different mix of people. Sex and age are confounders, and we can see them directly.
- Stratify on $\{$sex, age$\}$: four strata, and every one of them is occupied.

The four strata

Stratum	n	first class	survival, 1st	survival, other	difference
Adult male	1,667	175	32.6%	18.8%	13.7 pp
Adult female	425	144	97.2%	62.6%	34.6 pp
Male child	64	5	100.0%	40.7%	59.3 pp
Female child	45	1	100.0%	61.4%	38.6 pp

- Every within-stratum difference is **smaller than the** 35.4 **pp raw gap.** Confounding was inflating the estimate.
- Look at the composition: 76% of all passengers were adult males — the stratum with the *smallest* effect — but only 54% of first class.
- Note the last stratum: **one** first-class female child. Remember her; she returns shortly.

Three estimands, three answers

Same four differences. Only the weights change.

Weight on	Adult male	Adult female	Male child	Female child
ATE (population share)	0.757	0.193	0.029	0.020
ATT (treated share)	0.538	0.443	0.015	0.003
ATU (control share)	0.795	0.150	0.031	0.023

$$\widehat{ATE} = 19.6 \text{ pp}, \qquad \widehat{ATT} = 23.8 \text{ pp}, \qquad \widehat{ATU} = 18.9 \text{ pp}$$

- The raw 35.4 pp falls to roughly **20 pp** once composition is held fixed. Nearly half the headline gap was confounding.
- $ATT > ATE$ because first class was disproportionately adult *female* — the stratum with a large effect.

The curse of dimensionality

- Now delete one passenger: the single first-class female child.
- That stratum still exists — 44 girls travelled in other classes — but it now has **no treated unit**. The within-stratum difference is undefined.

Parameter	Still identified?
ATE	**No** — needs all four differences
ATU	**No** — female children get weight 0.023
ATT	**Yes** — 23.7 pp from the three occupied strata

- One passenger out of 2,201 destroys two of the three parameters. The *ATT* survives because it never needed that cell.
- This is common support failing.

Why it gets worse fast

- One binary covariate: 2 strata. Ten binary covariates: $2^{10} = 1{,}024$. One continuous covariate: infinitely many.
- Cells multiply exponentially; your sample does not. Empty cells become the norm, not the exception.
- Non-parametric methods — subclassification, exact matching, reweighting — **break down honestly**: they simply cannot compute the estimate.
- Regression **does not break down**. It extrapolates across the empty cells using functional form, and reports a number with a standard error.

Exact Matching

Matching as explicit imputation

- Every causal inference is an imputation problem: the counterfactual is missing and must be filled in.
- Most methods impute **implicitly**, inside a formula. Matching does it **explicitly**, unit by unit — and that transparency is its main virtue.
- Start from the definition of the ATT. Exactly **one** term is unavailable:

$$ATT = \underbrace{E[Y^1 \mid D{=}1]}_{\text{observed: } Y=Y^1 \text{ when } D=1} - \underbrace{E[Y^0 \mid D{=}1]}_{\textbf{missing}}$$

- The first term is just the treated sample mean. Everything matching does is build a stand-in for the second.

Replacing the missing term

Condition on X, and the assumption does its work:

$$E[Y^0 \mid D{=}1, X{=}x] = \underbrace{E[Y^0 \mid D{=}0, X{=}x]}_{\text{weak unconfoundedness}}$$

$$= \underbrace{E[Y \mid D{=}0, X{=}x]}_{\text{switching equation}}$$

$$E[Y^0 \mid D{=}1] = \underbrace{\sum_x E[Y \mid D{=}0, X{=}x]\,\Pr(X{=}x \mid D{=}1)}_{\text{common support: this must exist wherever treated units live}}$$

- The missing quantity is now written entirely in terms of things you can estimate from control units.
- **Matching is simply a non-parametric estimate of** $E[Y \mid D{=}0, X{=}X_i]$ — take the controls that look like i, and average them.

The estimator, and what to do about ties

Let $\mathcal{J}(i)$ be the set of control units matched to treated unit i, and $M_i = |\mathcal{J}(i)|$.

$$\widehat{ATT} = \frac{1}{N_T} \sum_{i\,:\,D_i=1} \Bigg(Y_i - \underbrace{\frac{1}{M_i} \sum_{j \in \mathcal{J}(i)} Y_j}_{\widehat{Y_i^0}} \Bigg)$$

- If two controls sit at exactly the same distance from i, use both — breaking the tie arbitrarily injects the analyst's coin flip into the estimate. <!– - M_i also varies when a caliper binds: a treated unit with no control inside the caliper contributes nothing, and N_T must then count only the units you actually matched.
- You can *point at the specific firms* doing the work for each observation. No regression output permits that. –>

The same idea for the ATE

For a control unit j, the missing half is Y^1. Let $\mathcal{I}(j)$ be the treated units matched to it, $M_j = |\mathcal{I}(j)|$.

$$\widehat{ATE} = \frac{1}{N}\Bigg[\underbrace{\sum_{i\,:\,D_i=1} \Big(Y_i - \frac{1}{M_i} \sum_{j \in \mathcal{J}(i)} Y_j \Big)}_{\text{treated: observed} - \text{imputed}} + \underbrace{\sum_{j\,:\,D_j=0} \Big(\frac{1}{M_j} \sum_{i \in \mathcal{I}(j)} Y_i - Y_j \Big)}_{\text{control: imputed} - \text{observed}} \Bigg]$$

- Every unit contributes one *estimated* treatment effect; the ATE averages all N of them.
- Equivalently $\widehat{ATE} = \frac{N_T}{N}\widehat{ATT} + \frac{N_C}{N}\widehat{ATU}$ — the weights from §4, back again.

Exact matching and its limit

- With discrete X and full common support, exact matching and subclassification give the **same answer**. They are the same estimator, organized differently.
- The difference is bookkeeping: subclassification aggregates to stratum effects first; matching imputes unit by unit.
- Same fatal problem, too: with a continuous covariate, **exact matches occur with probability zero.**
- Two responses, and they define the next two sections:
  - Give up on exact equality — match to the *nearest* unit (§6), then correct the resulting error (§8).
  - Give up on matching X itself — match on a *summary* of X (§9).

Nearest-Neighbour Matching

One confounder

- Drop the demand for exact equality. Match each treated unit to the control that minimizes $|X_i - X_j|$.
- The gap you accept is called the **matching discrepancy**, and §8 is entirely about what it costs you.
- Choices you must make and report:
  - **With or without replacement.** With replacement always takes the closest match, so it **lowers bias** and raises **variance.**
  - **How many neighbours,** M**.** More neighbours: less variance, more bias.
  - **Caliper**: refuse any match beyond a maximum distance. Better to drop a treated unit than to match it to something unlike it.

Multiple dimensions and the distance metric

With several covariates, "closest" requires a definition — and the choice changes the answer.

- **Euclidean**: $\sqrt{\sum_k (X_{ik} - X_{jk})^2}$. Scale-dependent: a variable in won swamps one measured as a ratio.
- **Normalized Euclidean**: divide each covariate by its standard deviation. Fixes scale, ignores correlation.
- **Mahalanobis**: $\sqrt{(X_i - X_j)'\Sigma^{-1}(X_i - X_j)}$. Standardizes scale **and** accounts for correlation between covariates. The default choice.

- Why correlation matters: if size and leverage are highly correlated, Euclidean distance double-counts what is essentially one dimension.

Inference with Matching

Why the usual standard errors fail

- Matched observations are **dependent by construction**: the same control may serve several treated units, and the matches were chosen *using the data*.
- Conventional standard errors treat the matched sample as if it were drawn at random. It was not, and they understate uncertainty.
- Abadie and Imbens (2008) show conventional standard errors is invalid for matching estimators
- Use the **Abadie–Imbens variance estimator**, which prices in how many times each control was reused.
- Practical note: check what your software reports *by default*. Several packages report the naive standard error unless told otherwise.

The Abadie–Imbens variance estimator

Let $\hat{\tau}_i = Y_i - \frac{1}{M}\sum_{m=1}^{M} Y_{j_m(i)}$ be unit i's **own** effect, so $\widehat{ATT} = \frac{1}{N_T}\sum_{i:D_i=1}\hat{\tau}_i$ is their average. K_j counts how often control j is used.

$$\hat{\sigma}^2_{ATT} \;=\; \underbrace{\frac{1}{N_T}\sum_{i\,:\,D_i=1}\left(\hat{\tau}_i - \widehat{ATT}\right)^2}_{\text{(a) spread of the unit-level effects}} \;+\; \underbrace{\frac{1}{N_T}\sum_{j\,:\,D_j=0}\frac{K_j(K_j-1)}{M^2}\,\hat{\sigma}^2(X_j)}_{\text{(b) penalty for reusing controls}}$$

$$\mathrm{SE}(\widehat{ATT}) \;=\; \sqrt{\hat{\sigma}^2_{ATT}\,/\,N_T}$$

- So term (a) is simply the **sample variance of the** $\hat{\tau}_i$. Alone, it is the without-replacement formula: with $K_j \in \{0,1\}$ term (b) vanishes.
- $\hat{\sigma}^2(X_j) = \mathrm{var}(Y \mid X_j, D{=}0)$, the outcome variance among controls.
- Here M is **fixed** — the clean $K_j(K_j{-}1)$ split needs it.

In Stata

Cunningham's training data: 10 trainees, 20 non-trainees, matched on age and gpa.

```
use https://github.com/scunning1975/mixtape/raw/master/training_inexact.dta, clear

* Minimized euclidean distance with usual variance and robust
teffects nnmatch (earnings age gpa) (treat), atet nn(1) metric(eucl) vce(iid)
teffects nnmatch (earnings age gpa) (treat), atet nn(1) metric(eucl) generate(match1)

* Minimized Mahalanobis distance with usual variance and robust
teffects nnmatch (earnings age gpa) (treat), atet nn(1) metric(maha) vce(iid)
teffects nnmatch (earnings age gpa) (treat), atet nn(1) metric(maha) generate(match2)
```

- `(earnings age gpa)` is outcome-then-covariates; `(treat)` is the treatment. `atet` asks for the ATT, `nn(1)` for one neighbour.
- `vce(iid)` is the constant-variance standard error. **Drop it and you get Abadie–Imbens** — that is Stata's default, so the second command of each pair is the one to report.
- `generate(match1)` writes each unit's matched partner back into the data. Always look at it.

Two metrics, two answers

	$\widehat{ATT}$	`vce(iid)`	**Abadie–Imbens**	distinct controls used
Euclidean	1607.5	38.7	38.7	10 of 20
Mahalanobis	1315.0	325.5	590.1	**3** of 20
Truth	1607.5			

- age has standard deviation 9.1 and gpa only 0.57, so raw Euclidean distance is **matching on age and ignoring gpa** — the scale problem from §6. Every trainee gets a *different* control: $K_j \leq 1$, term (b) is zero, both standard errors coincide.
- Mahalanobis makes gpa count — and gpa is noise. **Three controls end up serving all ten trainees**; one is used five times.
- Term (a) alone still reports 199.8. The reuse correction triples it.

Bias Correction

The matching discrepancy is bias, not noise

Matching was only *approximate*: $X_{j(i)} \approx X_i$, never $=$. What does the leftover gap cost?

- Abadie–Imbens (2006): More data does not help much.
- Usually you cannot get more data anyway. Abadie–Imbens (2011) is the second-best answer: **model the bias and subtract it.**

An outcome model for the missing Y^0

$$\mu^0(x) = E[\, Y \mid X{=}x, D{=}0\,] \;=\; \underbrace{E[\, Y^0 \mid X{=}x\,]}_{\text{unconfoundedness}}$$

$$\mu^1(x) = E[\, Y \mid X{=}x, D{=}1\,] \;=\; \underbrace{E[\, Y^1 \mid X{=}x\,]}_{\text{unconfoundedness}}$$

$$Y_i = \mu^{D_i}(X_i) + \varepsilon_i, \qquad E[\, \varepsilon_i \mid X_i, D_i\,] = 0$$

- The second equality on each line **is** unconfoundedness. It is what lets a regression fitted on controls speak about treated units.
- So far, all definition. The **assumption** arrives only when we choose a functional form for $\hat{\mu}^0$.

Where the bias hides

Substitute $Y_i = \mu^{D_i}(X_i) + \varepsilon_i$ into $\widehat{ATT} = \frac{1}{N_T}\sum_{D_i=1}(Y_i - Y_{j(i)})$, subtract the true ATT, then add and subtract $\frac{1}{N_T}\sum_{D_i=1}\mu^0(X_i)$:

$$\begin{aligned}\widehat{ATT} - ATT \;=\;& \underbrace{\frac{1}{N_T}\sum_{D_i=1}\left(\mu^1(X_i) - \mu^0(X_i) - ATT\right)}_{\text{(1) sampling variation in } \textit{which} \text{ units got treated}} \\ &+ \underbrace{\frac{1}{N_T}\sum_{D_i=1}\left(\varepsilon_i - \varepsilon_{j(i)}\right)}_{\text{(2) mean-zero noise}} \\ &+ \underbrace{\frac{1}{N_T}\sum_{D_i=1}\left(\mu^0(X_i) - \mu^0(X_{j(i)})\right)}_{\text{(3) }\textbf{matching bias}}\end{aligned}$$

- Scaled by $\sqrt{N_T}$, rows (1) and (2) converge to mean-zero normals. They are noise, and they average away.
- Row (3) does not. Its summands are **signed**: if the nearest control is systematically older, poorer, smaller, then every term carries the same sign.

Two things make the bias big

$$\text{Matching bias} \;=\; E\left[\, \mu^0(X_i) - \mu^0(X_{j(i)}) \,\middle|\, D = 1 \,\right]$$

- Read it as a product of two quantities, and it is zero if **either** is:
  - **How bad the matches are** — the size of $X_i - X_{j(i)}$. Perfect matching kills it.
  - **How steep μ^0 is** — the heterogeneity of Y^0 across strata of X. A flat μ^0 kills it too.
- Match a 13-year-old to a 15-year-old and nothing is lost *unless* Y^0 differs between 13- and 15-year-olds.
- Note which potential outcome appears. For the ATT it is Y^0 that is missing and imputed, so it is μ^0 — and only μ^0 — that has to be modelled.

The Abadie–Imbens procedure, in five steps

1. **Match.** Nearest neighbour on your metric; form $\hat{\delta}_i = Y_i - Y_{j(i)}$ for each treated unit.
2. **Regress.** OLS on the **matched control units only**: $Y_j = \alpha + \beta X_j + e_j$.
3. **Predict.** $\hat{\mu}^0(X_i) = \hat{\alpha} + \hat{\beta} X_i$ and $\hat{\mu}^0(X_{j(i)}) = \hat{\alpha} + \hat{\beta} X_{j(i)}$.
4. **Subtract** the estimated bias:

$$\widehat{ATT}^{BC} = \frac{1}{N_T} \sum_{D_i=1} \Big[(Y_i - Y_{j(i)}) - (\hat{\mu}^0(X_i) - \hat{\mu}^0(X_{j(i)})) \Big]$$

5. **Report an Abadie–Imbens standard error** (§7). The correction changes the estimate, not the reuse problem.

- Step 2 never touches a treated outcome: $\hat{\mu}^0$ is fitted where Y^0 is *observed*, then transplanted. Unconfoundedness is what permits the transplant.

Only the gap has to be right

- The estimator uses only the **difference** $\hat{\mu}^0(X_i) - \hat{\mu}^0(X_{j(i)})$. With one covariate and a linear fit that is simply $\hat{\beta}\,(X_i - X_{j(i)})$.
- So $\hat{\mu}^0$ must be accurate *over the matching gap*, not globally. Get the level wrong everywhere and the correction is unaffected — $\hat{\alpha}$ cancels.
- **This is the whole reason bias correction is more credible than regression alone.** Regression extrapolates $\hat{\mu}^0$ across the entire covariate space; bias correction only interpolates it across a gap that matching already made small.
- Division of labour: **matching does the heavy lifting on comparability; regression cleans up the remainder.**
- Corollary: the worse your matches, the more work the regression does, and the more the method degenerates into regression.

Ten workers

Unit	Worker	Y^1	Y^0 / $\widehat{Y^0}$	$\hat{\delta}_i$	Age	D
1	Andy	\$200	\$195	\$5	23	1
2	Betty	\$250	\$225	\$25	27	1
3	Chad	\$150	\$140	\$10	29	1
4	Doug	\$300	\$195	\$105	22	1
5	Edith	—	\$225		27	0
6	Fred	—	\$500		32	0
7	Gina	—	\$200		26	0
8	Hank	—	\$190		26	0
9	Inez	—	\$180		28	0
10	Janet	—	\$140		29	0

- Andy (23) and Doug (22) both match Gina and Hank, the 26-year-olds: $\widehat{Y^0} = \frac{200+190}{2} = 195$. Betty and Chad match exactly.
- $\widehat{ATT} = (5 + 25 + 10 + 105)/4 = \$36.25$. Fred and Inez are never used.
- Both inexact matches point the same way — the match is *older* than the trainee.

Steps 2–4, worked through

Step 2, over the four *matched* controls — Edith $(27, 225)$, Gina $(26, 200)$, Hank $(26, 190)$, Janet $(29, 140)$: $\hat{\mu}^0(\text{age}) = 683.75 - 18.33\,\text{age}$.

Worker	Age	Match age	$\hat{\delta}_i$	$\hat{\beta}(X_i - X_{j(i)})$	$\hat{\delta}_i^{BC}$
Andy	23	26	\$5	\$55.0	$-$\$50.0
Betty	27	27	\$25	\$0	\$25.0
Chad	29	29	\$10	\$0	\$10.0
Doug	22	26	\$105	\$73.3	\$31.7

$\widehat{ATT} = \$36.25 \longrightarrow \widehat{ATT}^{BC} = \$4.17.$

- Exact matches get **zero** correction by construction. Only the two bad matches move.
- $\hat{\beta} < 0$: among *these four* controls earnings fall with age. Refit on all six and Fred $(32, \$500)$ flips it to $\hat{\beta} = +43.7$, giving $\widehat{ATT}^{BC} = \$112.6$.
- Four points and one outlier. The "small" regression is doing an enormous amount of work — that is the method's exposure.

What the correction does and does not do

- It converts a *matching* problem into a *much easier regression* problem. It does not eliminate the regression.
- It is honest about its own assumption: you must name a functional form for $\hat{\mu}^0$, and the ten-worker example shows how much that choice can move the answer.
- It does **not** repair the matches.
- Practical rule: report the raw and the bias-corrected estimate side by side. A large gap between them is a **diagnostic** — it says your matches were poor and your outcome model is now load-bearing.

Propensity Scores

Defining the propensity score

- **Propensity score**: $e(X) = \Pr(D = 1 \mid X)$
  - "the conditional probability of assignment to a particular treatment given a vector of observed covariates"
- **Balancing property** (Rosenbaum–Rubin 1983): $X \perp D \mid \Pr(D = 1 \mid X)$
  - $E[X \mid D = 1, \Pr(D = 1 \mid X)] = E[X \mid D = 0, \Pr(D = 1 \mid X)]$
  - Covariate distributions are similar between treatment and control conditional the propensity score.
- **Propensity score theorem** (Rosenbaum–Rubin 1983): if $(Y^1, Y^0) \perp\!\!\!\perp D \mid X$, then

$$(Y^1, Y^0) \;\perp\!\!\!\perp\; D \;\mid\; e(X)$$

  - Conditioning on a **single number** does what conditioning on all of X does. Ten covariates collapse to one scalar — the curse of dimensionality defeated, for *conditioning*.

Estimating the score

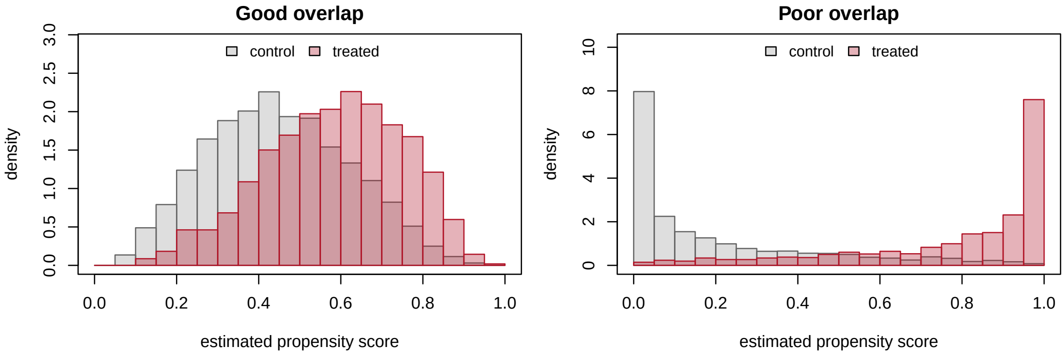
- You know the **true** score only when you ran the experiment yourself. Otherwise, it must be estimated: $\Pr(D_i{=}1 \mid X_i) = F(\beta X_i)$, fitted by **logit or probit**.
- Judge the specification by **balance achieved**, not by fit. A score with a high pseudo-R^2 but poor balance is a bad score; the reverse is fine.
- Fit, check balance, add interactions and polynomials where balance fails, refit. A legitimate specification search, because it never looks at Y — the whole stage is **blind to the outcome**, which protects against fishing. Say so in the paper.
- **The score creates no support that X lacks.** If a covariate cell is all-treated or all-control, collapsing X to a scalar hides that; it does not fix it.

Three things to do with the score

Once every unit has a $\hat{e}(X_i)$, there are three standard uses.

1. **Diagnose.** Plot $\hat{e}$ separately for treated and control units. This is the common-support check, and it costs nothing.
2. **Weight.** Reweight the sample to build a pseudo-experiment — inverse probability weighting.
3. **Match.** Pair treated to control units with similar scores, then difference — propensity score matching.

Visualizing overlap



Right panel: a treated mass at $\hat{e} \approx 1$ with no controls, and a control mass at $\hat{e} \approx 0$ with no treated. Those units have **no counterfactual**. Any estimate covering them is extrapolation, not comparison.

This plot belongs in every such paper

- Common support is verifiable, unlike unconfoundedness.
- What to do when it looks like the right panel:
  - **Trim** observations with extreme scores, or impose a caliper. Drop units with $\hat{e} \notin [0.1, 0.9]$. A rule of thumb, but a defensible and widely accepted one (Crump, Hotz, Imbens and Mitnik, 2009).
  - **Report the change in estimand.** After trimming you describe a subpopulation, not the original one. Say how many observations went and how the treated sample shifted.
  - Consider retreating to the ATT, which only needs overlap in one direction.
- What *not* to do: report the estimate and stay silent. The regression will happily extrapolate into a region with no data and give you a standard error that reflects none of that.

Inverse probability weighting

Rather than matching, **reweight**: treated get $1/\hat{e}(X)$, controls $1/(1-\hat{e}(X))$.

$$\widehat{ATE}_{IPW} = \frac{1}{n}\sum_i \frac{D_i Y_i}{\hat{e}(X_i)} - \frac{1}{n}\sum_i \frac{(1-D_i)Y_i}{1-\hat{e}(X_i)}$$

Why *that* weight? Take the first term, use $D\,Y = D\,Y^1$, and condition on X:

$$\begin{aligned} E\left[\frac{D\,Y}{e(X)}\right] &= E\left[\frac{E[D\,Y^1 \mid X]}{e(X)}\right] = E\left[\frac{E[D \mid X]\,E[Y^1 \mid X]}{e(X)}\right] \\ &= E\big[\,E[Y^1 \mid X]\,\big] = E[Y^1] \end{aligned}$$

- Step 1 is the **switching equation**; step 2 **is unconfoundedness**
- It is the unique weight that undoes selection (Horvitz–Thompson 1952; Imbens 2000).

Propensity score matching

- **Use 3**: match each treated unit to controls with a **similar score**, then difference. All the §6 machinery applies directly.
- **Caliper** matching: refuse any match farther than some tolerance (say 0.001 on the score), *discarding* the treated unit if none qualifies. **Radius** matching: average all controls inside the tolerance.
- Two problems with both, neither of which shows up in a standard error:
  - The tolerance is **chosen by the researcher**, with no theoretical rule (Huntington-Klein et al. 2021).
  - Dropping treated units **changes the estimand** — the ATT summarised the *original* treated sample. Cunningham: caliper matching "ends up also shifting the goalposts in unknown ways."
- The score is estimated, so its sampling error enters the standard errors (Abadie–Imbens 2016).
- The caveat that matters most: same score, different X. **The three cells** — a treated Large/Low-leverage firm from A, a control Small/High-leverage firm from B, both at $e = 0.2$ — and the algorithm pairs them.

What the score does and does not buy

- It buys **dimension reduction**, not identification. If X omits a confounder, $e(X)$ omits it too.
- You have swapped an assumption about the **outcome** model for one about the **treatment** model — a choice about which model you would rather defend. The next slides show you can decline to choose.
- Its most valuable use may be **diagnostic**: the overlap plot reveals problems that no regression table displays.
- Read the fit **backwards**. A treatment model that predicts D *badly* is good news — $\hat{e}$ is nearly flat, so treatment is close to as-if random given X, and everyone has a counterpart.
- The alarm is a model that predicts D almost perfectly: it drives scores to 0 and 1, and leaves nobody to compare with. High pseudo-R^2 in the treatment equation is a warning, not an achievement.

Regression

Regression as a conditioning strategy

- Regression belongs in this chapter. It is not a rival to matching — it is another way of executing the same substitution from §3.
- Under unconfoundedness and **constant** treatment effects, $Y = \alpha + \beta D + X'\gamma + \varepsilon$ recovers β, because unconfoundedness delivers exactly the exogeneity condition OLS needs:
$$(Y^1, Y^0) \perp\!\!\!\perp D \mid X \implies E[\varepsilon \mid D, X] = 0$$
- Recall Lecture 1: the error term contains Y^0. Selection bias was $\mathrm{cov}(D, Y^0) \neq 0$. Conditioning on X is the attempt to make that covariance zero *within strata*.
- What regression adds that matching lacks: it **interpolates and extrapolates** with functional form, so empty cells do not stop it.
- What it hides: it does that silently.

Frisch–Waugh–Lovell

The coefficient on D in a multiple regression equals the coefficient from a **bivariate** regression of residualized Y on residualized D:

```
set.seed(975)
n <- 20000; X <- rnorm(n); D <- rbinom(n, 1, plogis(1.2 * X))
Y <- 2 * D + 3 * X + rnorm(n)
c(direct = unname(coef(lm(Y ~ D + X))[2]),
  fwl    = unname(coef(lm(resid(lm(Y ~ X)) ~ resid(lm(D ~ X))))[2]))
```

```
  direct      fwl
2.018086 2.018086
```

- Identical to machine precision. "Controlling for X" *means* "using only the variation in D that X cannot explain."
- This is the cleanest statement of what a control variable does — and it tells you where the identifying variation comes from.
- It also warns you: if X explains nearly all of D, almost no variation remains, and the estimate rests on very few effective observations.

What FWL implies about weights

- FWL says OLS uses the residual variation in D. Where is that variation largest? Where treatment is **least predictable**.
- Formally, with X dummies, OLS weights each stratum by $N_j\, p_j(1 - p_j)$ — the **conditional variance of treatment** within it.
- So OLS gives most weight to strata where treatment is closest to a coin flip, and least where it is nearly deterministic.
- That is a defensible **efficiency** motive: those strata carry the most information per observation.
- It also means **OLS is not estimating the** ATT — or the ATE — when effects are heterogeneous. It estimates a variance-weighted average with no policy interpretation.

Regression adjustment (imputation)

- A more flexible use of regression that mirrors matching directly:
  1. Fit $\hat{\mu}^1(x)$ on the **treated** and $\hat{\mu}^0(x)$ on the **controls**, separately.
  2. For *every* unit, impute both potential outcomes and average the difference:

$$\widehat{ATE} = \frac{1}{n} \sum_i \left(\hat{\mu}^1(X_i) - \hat{\mu}^0(X_i) \right)$$

- Separate models allow the covariates to matter differently under treatment and control — exactly what a single β on D forbids.
- This is the same imputation logic as matching, with a model in place of a neighbour.
- Combining it with weighting gives **doubly robust** estimators: consistent if *either* the outcome model or the propensity model is right. A genuinely useful hedge, and increasingly the default in applied work.

Matching versus regression

- Both rely on the **same** identifying assumption: unconfoundedness. Neither creates exogenous variation.
- The difference is in *how they weight strata* when the effect is heterogeneous.
  - **Matching (ATT)** weights stratum j by the number of **treated** units in it.
  - **OLS with X dummies** weights stratum j by $N_j \, p_j(1 - p_j)$.
- They also differ in what they do about missing cells: matching refuses; regression extrapolates.
- So when the two disagree, ask two questions in order: *is it the weights, or is it the extrapolation?* Both are diagnosable.

The same example under both

Two strata. $p_{\text{small}} = 0.5$, $p_{\text{large}} = 0.1$; effects $+2$ and $+3$; 100 firms per stratum.

	Weight on small	Weight on large	Estimate
Matching (ATT): treated counts	$50/60 = 0.833$	$10/60 = 0.167$	**2.167**
OLS: $N_j p_j(1 - p_j)$	$25/34 = 0.735$	$9/34 = 0.265$	**2.265**
ATE: population shares	0.5	0.5	**2.500**

```
X <- c(rep(0, 100), rep(1, 100))
D <- c(rep(c(1, 0), each = 50), rep(c(1, 0), times = c(10, 90)))
Y <- ifelse(X == 0, 0, 5) + ifelse(X == 0, 2, 3) * D
round(unname(coef(lm(Y ~ D + factor(X)))[2]), 3)
```

```
[1] 2.265
```

Reading the comparison

- The regression returns 2.265 — exactly the variance-weighted average, not the ATT and not the ATE.
- The gap is small here. Angrist and Pischke argue it usually is, in practice.
- But "usually small" is not "zero", and the difference is entirely about **which firms your estimate describes**:
  - Matching describes the treated.
  - OLS describes wherever treatment was most unpredictable.
- **When it matters most**: when treatment probability varies a lot across strata *and* effects are heterogeneous. Both are typical in finance.
- This is the same variance-weighting issue that resurfaces, far more destructively, in staggered difference-in-differences.

What Matching Cannot Do

Matching creates no new variation

- The most important slide in this lecture.
- Matching **does not introduce exogenous variation**. It reorganizes and reweights the data you already have, using observables you already measured.
- Consequence: if your OLS regression was biased by endogeneity, **the matched estimate is biased too** — typically by a similar amount, for the same reason.
- Contrast this with the designs still to come:
  - **IV** imports variation from outside the model.
  - **RDD** exploits assignment that is locally as good as random.
  - **DiD** differences out fixed unobservables.
- Matching belongs to a different category. It is a *conditioning strategy*, like regression — not a research design.

What it cannot fix

Matching is powerless against:

- **Unobserved confounders.** You cannot match on what you cannot measure.
- **Reverse causality.** Matching on X says nothing about whether Y caused D.
- **Measurement error** in treatment or outcome.
- **Bad controls.** Matching on a post-treatment variable is exactly as damaging as controlling for it — Lecture 2 applies unchanged, and §2's collider F is the warning.

You cannot match on what you cannot measure.

- A matched sample looks reassuringly balanced on the covariate table — and that reassurance is *precisely* the danger. Balance on observables is silent about unobservables.

So what is it good for?

Given all that, matching still earns its place:

1. **A non-parametric benchmark.** It estimates the effect without assuming a functional form. If matching and OLS disagree sharply, your specification is doing more work than you thought.
2. **Common-support discipline.** Matching refuses to compare units with no counterpart. Regression silently extrapolates into regions with no data.
3. **Diagnostic value.** The propensity score distribution reveals overlap problems that a regression table hides completely.
4. **Robustness when the functional form is wrong.** Matching can outperform a misspecified regression in finite samples.

- Use it as a *complement* to regression and a check on it — not as an identification strategy.

The NSW Demonstration

The 1970s empirical crisis

- The methods in this lecture were not adopted because they were elegant. They were adopted after a crisis of confidence in observational work.
- Job-training programs in the 1970s were evaluated observationally, and the estimates were all over the place — depending, apparently, on the analyst.
- The **National Supported Work (NSW) Demonstration** was different: it *randomized* participation. So the experimental effect was known.
- That created something rare and valuable: a **benchmark**. You could now ask whether observational methods, applied to the same treated units, recover the experimental answer.
- Everything in this lecture gets graded against it.

What LaLonde did

- LaLonde (1986) ran the decisive test. He took the NSW experiment, **threw away the experimental control group**, and replaced it with comparison groups built from survey data (CPS and PSID).
- He then applied the standard observational toolkit of the day and compared the results to the experimental benchmark of roughly **$1,800** in annual earnings gains.
- The observational estimates ranged from a fraction of the benchmark to **the wrong sign** — and which one you got depended heavily on how the comparison group was built.
- The paper was devastating and hugely influential. It is a major reason the profession moved toward IV, RDD, and DiD — the other river.

Dehejia and Wahba: not so fast

- **Dehejia and Wahba (1999, 2002)** revisited LaLonde's data with propensity score methods.
- Their finding: the observational estimates *did* recover the experimental benchmark reasonably well — **once common support was imposed**.
- The key diagnosis: much of LaLonde's failure was an **overlap** problem, not only a confounding problem. The CPS and PSID comparison groups contained enormous numbers of people utterly unlike the NSW participants.
- Discard the non-comparable units and the methods perform far better.
- This is why §3's distinction matters so much. Two different assumptions were failing, and only one of them is checkable.

Smith and Todd: not so fast either

- **Smith and Todd (2005)** re-examined the Dehejia–Wahba result and found it fragile.
- The success was **sensitive to the subsample** analysed and to the **propensity score specification** chosen. Reasonable alternative choices undid it.
- Their charge is uncomfortable and worth stating plainly: the specification that worked may have been found *because* the experimental answer was already known.
- You cannot run that search in real research. In a genuine application there is no benchmark to check against — that is the entire point of doing the study.

The honest summary

- Selection on observables **can** work — when overlap is genuine and the confounders are truly measured.
- Neither condition is verifiable in the setting where you need them:
  - Unconfoundedness is **untestable**, always.
  - Common support is **checkable**, and checking it was most of what rescued Dehejia–Wahba.
- So the practical lesson is asymmetric: do the check you can do, and argue explicitly for the assumption you cannot.
- **A useful exercise**: the NSW data are freely available. Apply every method in this lecture and compare them to the $1,800 benchmark. Cunningham's chapter walks through it, and it is the fastest way to develop instincts about when these methods hold up.

Wrapping Up

Summary [Part 1]

- **Assignment mechanisms resolve bias, not models.** The SDO decomposition is an identity; only a claim about how treatment was assigned deletes its bias terms.
- **Covariates come in four kinds**: confounders (include), colliders (never), precision covariates (optional), mediators (never). Pre-treatment timing is your best protection.
- **Unconfoundedness**, $(Y^1, Y^0) \perp\!\!\!\perp D \mid X$: randomization *within* strata of X. Untestable; defended with institutional knowledge, and Imbens's third defence is the strongest in finance.
- **Common support** is a *separate* assumption — and unlike unconfoundedness, it is **checkable**. Plot the propensity scores.
- **Subclassification** makes the estimand a choice about weights. On the Titanic: $SDO = 35.4$ pp, but $ATE = 19.6$, $ATT = 23.8$, $ATU = 18.9$.
- The **curse of dimensionality** is structural, not a small-sample nuisance. One missing passenger killed two of three parameters.

Summary [Part 2]

- **Matching** imputes counterfactuals explicitly. The matching discrepancy is systematically signed, so it becomes **bias** that shrinks slower than the standard error — hence bias correction.
- **Propensity scores** reduce X to one number and defeat dimensionality for *conditioning* — but not for identification. Judge the score by balance, not fit.
- **IPW** reweights to a pseudo-experiment; watch for exploding weights as $\hat{e} \to 0, 1$.
- **Regression** is a conditioning strategy too. FWL shows it uses the residual variation in D, which is why it weights by $N_j p_j(1 - p_j)$: 2.167 versus 2.265 in our example.
- **Matching creates no new exogenous variation.** If OLS was endogenous, matching is too.
- **NSW**: these methods can work when overlap is real and confounders are measured — and you can only verify one of those.
- Next lecture: designs that *do* create variation — starting with **regression discontinuity**.

A checklist for your own paper

1. State the **estimand** before the estimator. ATE, ATT, or ATU — and say why the policy question needs that one.
2. Defend the **assignment mechanism** in institutional terms. Name the confounders and say how you measured them.
3. Justify every covariate as **pre-treatment**. Any post-treatment control needs an argument, not a footnote.
4. **Plot the overlap.** Report what you trimmed and how the estimand changed.
5. Report **matching and regression**. If they disagree, diagnose whether it is weights or extrapolation.
6. Run a **placebo** on a pre-treatment period.
7. Say plainly what would break the design — and what unobservable would have to exist.

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